

Transience of the 3D Simple Random Walk Using Borel–Cantelli and Stirling

Let $S_n = X_1 + \dots + X_n$ be the simple symmetric random walk on $\mathbb{Z}^3$. We prove transience by showing $\sum P(S_n = 0) < \infty$ and then applying the first Borel–Cantelli lemma.

Step 1: Borel–Cantelli Reduction

Let $A_n = \{S_n = 0\}$. If $\sum P(A_n) < \infty$, then by the first Borel–Cantelli lemma, $P(A_n \text{ infinitely often}) = 0$. Thus it suffices to show $\sum P(S_n = 0) < \infty$.

Step 2: Count Paths Returning to the Origin

For $2n$ steps, let $2a, 2b, 2c$ be the numbers of steps in the x -, y -, and z -directions, with $a+b+c=n$. The number of return paths is $(2n)! / [(2a)!(2b)!(2c)!] \times C(2a, a)C(2b, b)C(2c, c)$. Therefore $P(S_{2n} = 0) = 6^{-n} \sum_{a+b+c=n} \frac{1}{a!^2 b!^2 c!^2}$.

Step 3: Bound the Sum

Let $T_n = \sum_{a+b+c=n} \frac{1}{a!^2 b!^2 c!^2}$. There are at most $(n+1)^2$ triples (a, b, c) . By convexity of $\log(m!)$, the maximum term occurs when a, b, c are as equal as possible. Hence $T_n \leq C n^2 / ((n/3)!)^6$.

Step 4: Apply Stirling's Formula

Using $m! \sim \sqrt{2\pi m} (m/e)^m$, $(2n)! \sim \sqrt{4\pi n} (2n/e)^{2n}$, $((n/3)!)^6 \sim n^3 (n/(3e))^{2n}$. Substituting yields $P(S_{2n} = 0) \leq C n^{-3/2}$.

Step 5: Summability and Conclusion

Since $\sum n^{-3/2} < \infty$, we have $\sum P(S_n = 0) < \infty$. By the first Borel–Cantelli lemma, $P(S_n = 0 \text{ infinitely often}) = 0$. Therefore the simple symmetric random walk on $\mathbb{Z}^3$ is transient (not recurrent).